Hashmaps

A hashmap (also known as a Hashtable) is a data structure that stores key-value pairs

It allows fast retrieval of values based on the keys by using a process called 'Hashing'.

The key is transformed into a unique index through a hash function, which can be used to access the value. (in constant time)

Real world example

→ Dictionary
→ Phone contacts — name of contact
→ Student ID system — Student ID

Practical Examples

→ Caching in web browser — hashmap to store cache of images etc.
→ DNS lookup
→ No SQL DBs (Redis, mongoDB)
→ load Balancer in distributed system
→ Compiler Symbol table

Why Hashing?

example of counting letter in a sentence

⇓

Counting words in a sentence.

→ how in an array key is fixed (int only)

control the key.

(want to decide the key type and the value type)

Implementation

1. Linked list — have to search as well ~~as do~~ no adding at need
2. BST
   → Some order & then do the same
3. Hashtable ⟶ O(1)

BST and hastable ~~using~~ used internally.

5 Question on Hashmap using dictionary

Show how the stack overflow Question solutions that

Explain properly

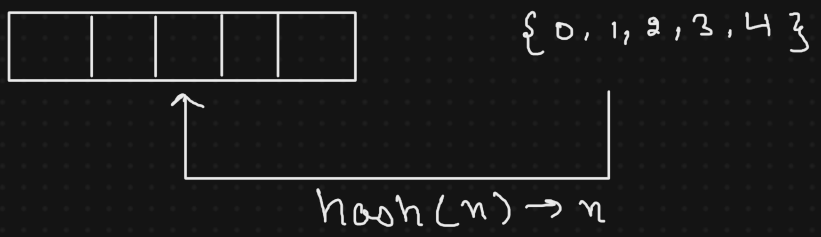
also on how list cannot be used to create Key

hash( )

Hashing

first show how in array or list the indexing helps in quick show/check

Index & item search ⇒ O(1) operation. free of ~~Del~~ len



but what for {0, 1, 2, 3, 1000}

⇒ memory wastage not efficient